SYEDA ZAINAB FATIMA

Computer Systems Engineer

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SUMMARY

Computer & Information Systems Engineering graduate with hands-on experience in machine learning, computer vision, and database-driven application development. Skilled in Python, and full-stack web technologies, with a track record of academic and independent projects spanning AI, network security, and software engineering. Seeking an IT/AI-related role to build reliable, scalable software solutions.

PROFESSIONAL EXPERIENCE

Intern — National Center in Big Data and Cloud Computing (NCBC), NED University Jun 2025 – Aug 2025

- Conducted research in computer vision, focusing on image processing and analysis.
- Handled and decoded metadata from XML, IMG, and TIFF files for large-scale datasets.
- Applied data preprocessing and feature extraction techniques to support research objectives.
- Collaborated with the research team to implement practical solutions in big data and cloud-based environments.

EDUCATION

Bachelor of Engineering in Computer Systems Engineering — NED University Oct 2022 – Aug 2026

GPA: 3.73

A-Levels in Computer Science — Credo College Aug 2020 – Jul 2022

CERTIFICATES & AWARDS

- Introduction to AI — Google (Coursera)
- Merit Scholarship for Academic Excellence — Credo College

SKILLS

- Python
- Machine Learning
- Computer Vision
- LangChain / LangGraph
- SQL / MySQL
- Django
- React.js
- Node.js
- HTML / CSS / JavaScript
- Microsoft Office

Core strengths: Adaptability, problem-solving, time management, team collaboration, and strong work ethic.

PROJECTS

Final Year Project – Logs Anomaly Detection System for Network Infrastructure

Client project for Ephlux

- Built a real-time network log anomaly detection system for different servers (Windows / Linux / Apache), using an ML ensemble to classify HTTP traffic into six attack types.
- Integrated live log monitoring with a Swift-based dashboard for real-time visibility into detected threats.

Object Detection – Vegetable Detection System

- Built a custom vegetable detection system using YOLO and DETR models, trained on a self-created dataset (manually collected images labeled with CVAT.ai).

AI Hand Gesture Recognition System

- Developed a gesture recognition system using a Convolutional Neural Network (CNN) to classify hand gestures in real time.

Distributed File Storage System

- Designed a fault-tolerant storage system implementing file chunking, replication, and load balancing across nodes.

Hotel Management System

- Built a full-stack hotel management application with Node.js and MongoDB on the backend and React.js on the frontend.

Pharmacy Management System

- Developed a pharmacy management system using Django and MySQL for the backend, with HTML, CSS, and JavaScript on the frontend.

Integrated Environmental Monitoring System

- Used C to fetch real-time weather data from a public API, analyze it, and generate automated reports.

Traffic Light Control System

- Implemented a traffic light controller using a Finite State Machine (FSM) as part of a hardware/digital logic project.

Online Banking System

- Collaborated with peers to build a secure, user-friendly online banking application.

Bake Shop Program

- Developed a bakery ordering application applying object-oriented programming (OOP) concepts.

Hangman Game

- Built a Hangman game using procedural programming concepts.